

CANDIDATE NAME (CG) _____

INDEX NUMBER _____



SERANGOON JUNIOR COLLEGE
JC2 Preliminary Examination 2013

BIOLOGY
Higher 2

Paper 3

9648
26 August 2013 / Monday
2 hours

Additional materials:
Answer paper

READ THESE INSTRUCTIONS FIRST

Write your name and index number in the spaces at the top of this page and on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions in all the sections.

FOR EXAMINER'S USE	
1	/11
2	/4
3	/15
4	/10
Planning	/12
Essay	/20
TOTAL	/72

At the end of examination,

1. Fasten all your work securely together;

INFORMATION FOR CANDIDATES

The intended number of marks is given in brackets [] at the end of each question or part question.

This question paper consists of 17 printed pages and 1 blank page

Question 1

a) Restriction fragment length polymorphism, RFLP, refers to differences in fragment length generated in different individuals by restriction enzymes.

Figure 1.1 below shows an autosomal dominant disease segregating in a family. Genomic DNA from each family member was assayed for RFLP for *EcoRI* sites, on Southern blots using probes derived from the gene.

The results for all individuals, except those for the foetus, are shown below the pedigree chart. The sizes of the DNA fragments obtained are as indicated.

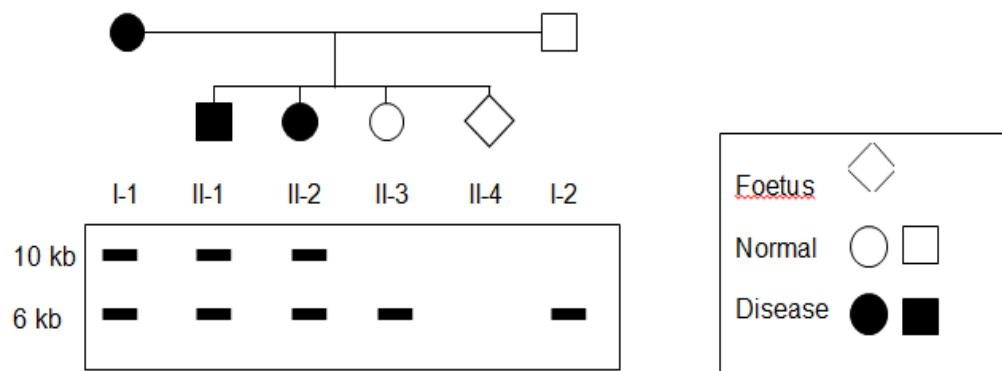
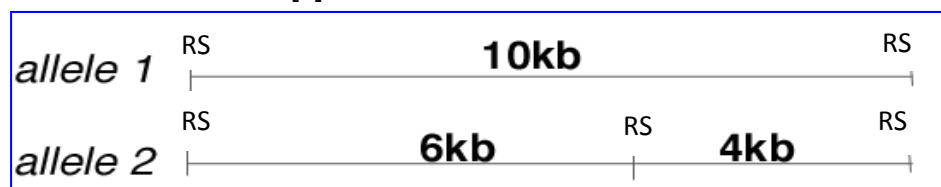


Figure 1.1

- i) The diagram below shows the 2 possible alleles of the gene. Indicate with thick straight lines on the diagram the possible location recognized by the probe that was added. [1]



RS= restriction site

- ii) Which RFLP fragment did the mother (I-1) contribute to the diseased children? [1]

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- iii) What is the probability of the foetus having the disease? Explain your answer. [2]

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- b) A different study was carried out on another family. **Figure 1.2** below shows a Southern blot with an RFLP detected by using a probe for a sequence located on a sex chromosome. The variant exhibits two forms, one at 8 kb and one at 6 kb. The mother, father and daughter are normal.

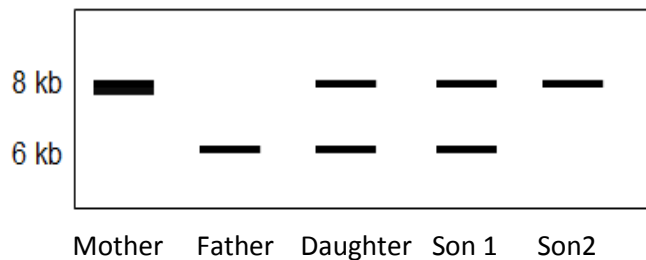


Figure 1.2

- i) From **Figure 1.2**, deduce which son is abnormal and explain how you arrive at your answer. [3]

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c) RFLP is widely used as a tool for disease detection. In disease detection, genetic markers are often used to predict if a person is likely to suffer from a particular disease.

i) Explain the features of genetic markers that allowed them to be used for the indirect analysis of the inheritance of disease. [2]

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d) DNA fingerprinting through the use of RFLP is a technique employed by forensic scientists to assist in the identification of individuals by their respective DNA profiles.

i) Explain why genes, especially protein-encoding genes, are seldom used for genetic fingerprinting. [2]

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[Total: 11 marks]

Question 2

Flavr Savr is a genetically modified tomato. It is made more rot-resistant by adding an antisense gene to interfere with the production of the enzyme polygalacturonase (PG). This enzyme causes the softening of the fruit by degrading pectin.

PG production during fruit ripening was measured in plants with one and with two antisense genes, and in normal plants. The results are shown in **Figure 2.2**.

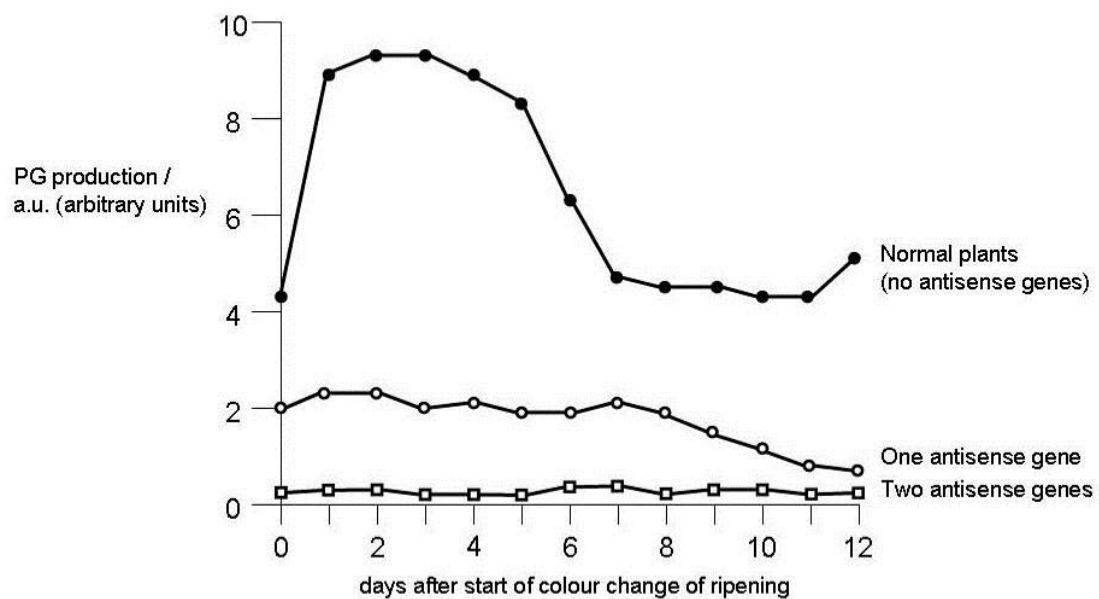


Figure 2.2

i) With reference to **Figure 2.2**, describe and explain the difference between normal tomato plants and transformed plant with one anti-sense gene. [4]

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[Total: 4 marks]

Question 3

a) Cystic Fibrosis (CF) is an autosomal recessive genetic disorder that affects predominantly the lungs and other mucosal organs in the body. A technique that measures the nasal potential difference has been investigated recently as a possible diagnostic tool. When chloride ion channels open, a potential difference will be recorded as a positive spike in the potential difference against time graph.

In the lab, laboratory researchers performed experiments on two groups of nasal epithelial cells, X and Y. To observe changes in nasal potential difference, the researchers perfused a solution of amiloride (Am) into the nasal activity, followed by the subsequent perfusion of the nasal cavity with a low chloride solution (LC).

Amiloride serves to block sodium absorption, resulting in a reduction of the potential difference. This provides an electrical gradient for the secretion of chloride, which is stimulated by the biochemical gradient provided by a low chloride solution.

The potential difference against time graph was plotted as shown in **Figure 3.1** below.

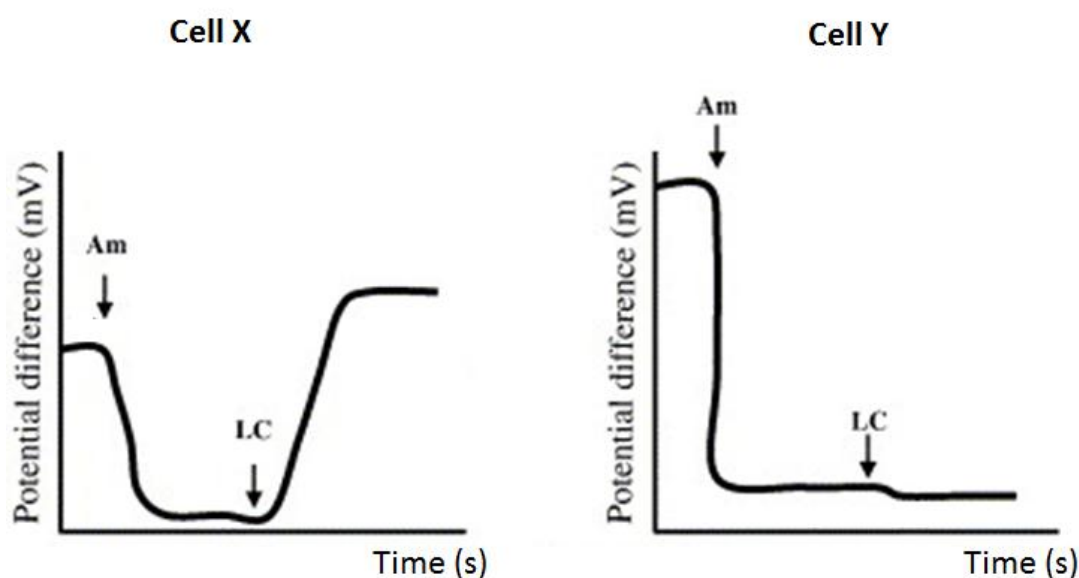


Figure 3.1

i) Explain what is meant by “autosomal recessive” genetic disorder. [2]

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- ii) With reference to **Figure 3.1**, identify the cell with cystic fibrosis. Explain your answer. [3]

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- iii) Describe how cystic fibrosis can be cured by gene therapy using a non-viral vector. [4]

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- iv) Suggest two technical challenges preventing the effectiveness of gene therapy for cystic fibrosis. [2]

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b) The procedure of current gene therapy for severe combined immunodeficiency disease (SCID) is illustrated in **Figure 3.2** below.

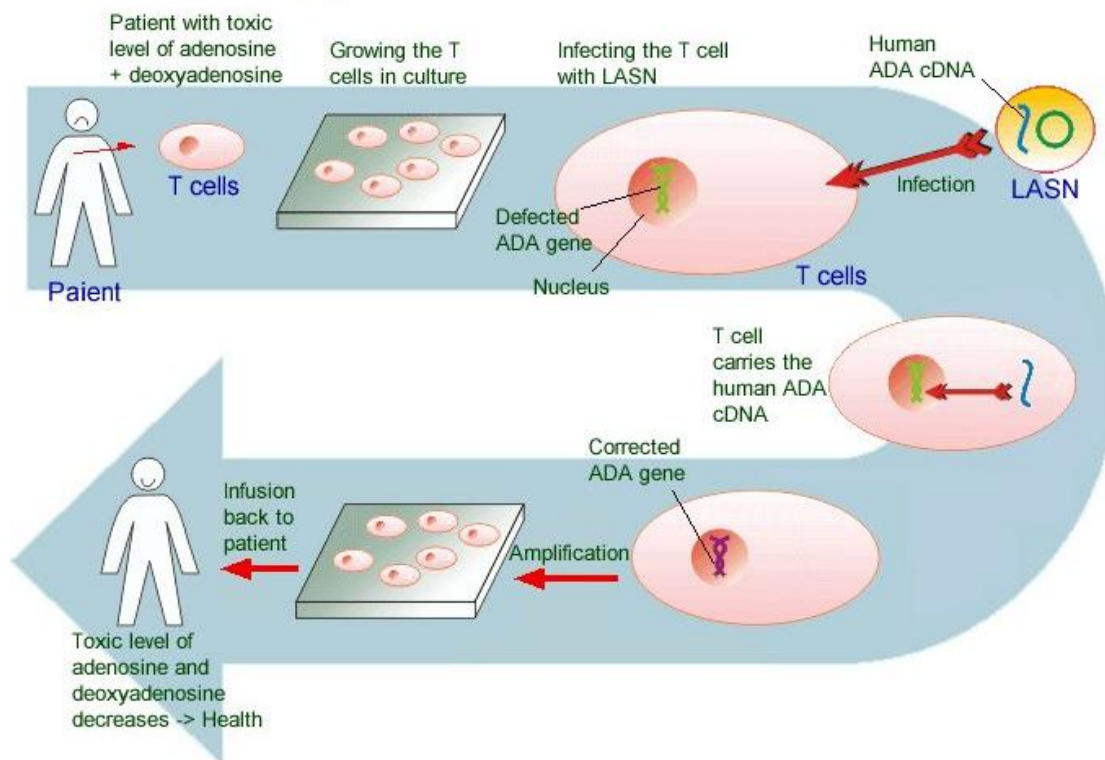


Figure 3.2

i) State two advantages of using the ex vivo method for gene therapy that is illustrated in **Figure 3.2**. [2]

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ii) Suggest and explain how you can improve the above gene therapy procedure in order to sustain long-term success of the treatment. [2]

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[Total: 15 marks]

Question 4

a) There has been rampant speculation that the consumption of genetically modified (GM) crop and livestock has impacted birth rates in the USA negatively. A study was carried out in the USA to investigate the claim. **Figure 4.1** shows a graphical plot of the data obtained.

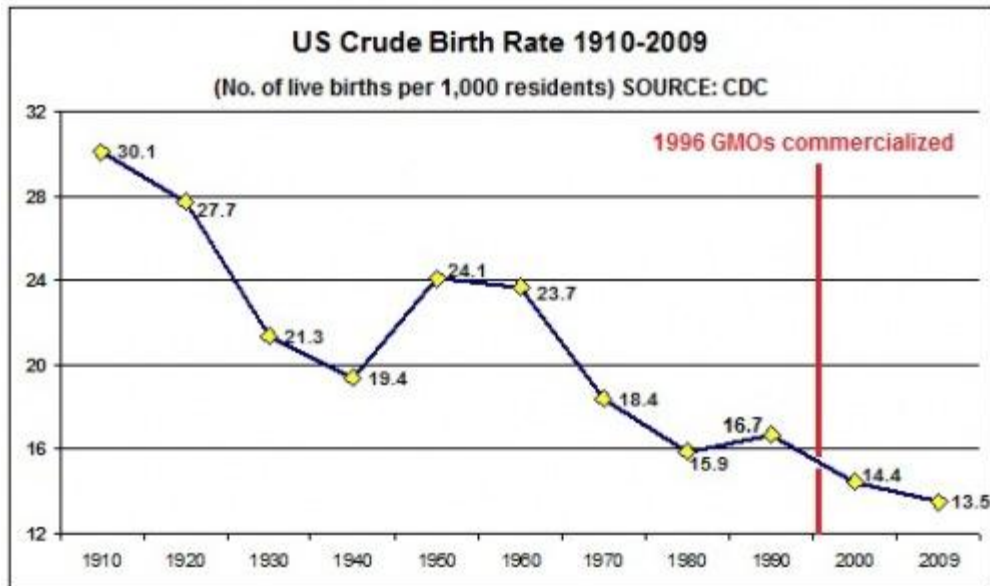


Figure 4.1

- i) With reference to **Figure 4.1**, comment whether the speculation can be justified by the data. [3]

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- ii) Describe two threats that GM food can pose to human health. [2]

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b) The effectiveness of CpTi against insect pests was tested using two groups of tobacco plants.

- Plants genetically engineered to express the allele coding for CpTi.
- Control plants which had been genetically engineered by inserting the allele coding for CpTi in the wrong orientation. These plants did not produce CpTi.

Equal numbers of each type of plant were infested with equal numbers of newly hatched larvae of an insect pest and left for seven days.

The mean percentage of leaf area eaten and the mean mass of surviving insects were then found. The results are shown in **Figure 4.2**.

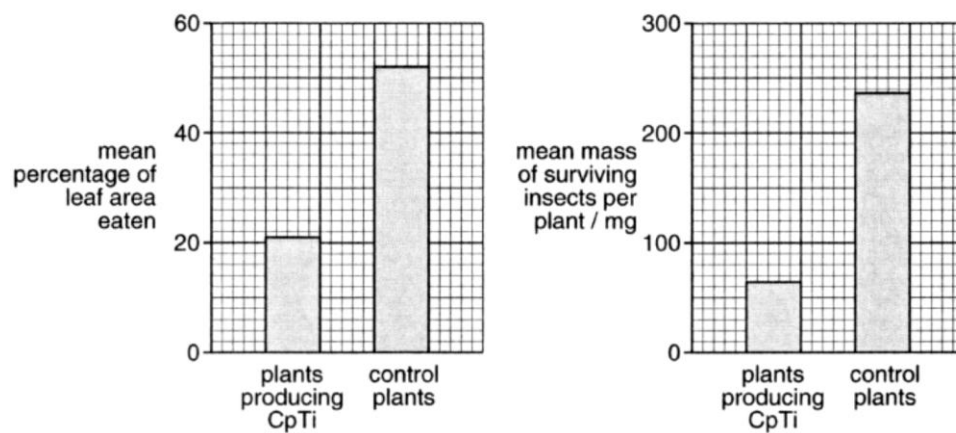


Figure 4.2

With reference to **Figure 4.2**,

i) Describe the effect of CpTi on the yield of tobacco leaves. [2]

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- ii) CpTi is a trypsin inhibitor. Suggest an explanation of the effect of the trypsin inhibitor, CpTi, on insects.[3]

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[Total: 10 marks]

Planning Question (12m)

A millionaire has recently passed on and news of his death has become the talk of the town. The deceased old man was known to be childless and thus his massive fortune amounting \$100 million was willed and donated to help the less fortunate orphans at Chen Su Lan Childrens' Home.

However, three days after his death, a man named Bernard Soon has stepped forward and claimed to be his son. According to Bernard Soon's allegations, the deceased millionaire had an affair with his mother during a year-long work trip in the 1930s. However, both had lost contact over the years and had not met since. Mr Bernard Soon has claimed that he has a picture of the millionaire and his mother, as well as his mother's ring with the millionaires' initials engraved onto it. Due to the large amount of money involved, this case has been covered at length by the media.

Bernard Soon has hired you as the lead researcher to prove his identity. Plan an investigation using RFLP analyses to confirm that Bernard Soon is indeed the son of the millionaire.

Your planning must be based on the assumption that you have been provided the following equipment and materials:

- Tissue sample from the millionaire and Bernard Soon
- Pestle and mortar
- DNA extraction buffer solution
- Microcentrifuge tubes
- Centrifuge
- Restriction enzymes
- Agarose or polyacrylamide gel plate
- Suitable source of electric current
- Radioactive probe
- Nitrocellulose membrane
- Autoradiography equipment

Your plan should have a clear and helpful structure to include:

- An explanation of the theory to support your practical procedure
- A description of the method used including scientific reasoning behind the method
- The type of data generated by the experiment
- How the results will be analysed including how the origin of the organism can be determined

This image shows a full page of a worksheet designed for handwriting practice. It consists of approximately 20 horizontal rows. Each row is defined by two parallel dashed lines, creating a series of uniform gaps for writing. The lines are evenly spaced across the entire page, providing a guide for letter height and placement. There is no text or other markings on the page.

[illegible]

[illegible]

[illegible]

Free-response question (20 marks)

Write your answer on the separate answer paper provided.

Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.

- a) Outline the large scale production of a **named commercially** important protein by genetic engineering. [10]
- b) Using specific example(s), compare viral and non-viral delivery systems in the delivery of a functional CFTR allele.[6]
- c) Describe the advantages and limitations of the Polymerase Chain Reaction (PCR). [4]

END OF PAPER

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